# NING YU

TACTILE PERCEPTION & APPLICATION · REINFORCEMENT LEARNING · IMITATION LEARNING · VLA & WM · SYSTEM IDENTIFICATION

Phil_ning@mail.dlut.edu.cn , 197-4110-3867 , Shanghai , github.com/NY-WaKeUp

## EDUCATION & TRAINING

**Dalian University of Technology (985)** 2024.09 – 2027.06

M.S. · Computer Science and Technology | Advisor: Prof. Liu Qian

· Dexterous-hand tactile perception and manipulation, reinforcement learning and imitation learning.

**Hebei University of Technology (211)** 2020.09 – 2024.06

B.Eng. · Internet of Things Engineering

## PROFESSIONAL EXPERIENCE

**AgiBot** 2026.07 – Present

Robot Motion Intelligence Intern Shanghai

- **Project 1 · Real-robot RL** Built a **HIL-SERL** real-robot RL pipeline for dexterous manipulation (O10hand + Franka Research 3); profiled the initial demo data to set the initial exploration-domain bounds, then grew the bounds sinusoidally during training, reaching **99%** success within **20 min**
- **Project 2 · real2sim2real & sim2sim** Built the RL training pipeline for the company flagship high-DoF hand; analyzed reward dynamics for **long-horizon credit assignment**; delivered simulation setup, training, sim-to-sim transfer, **system identification** and **policy deployment**
- **Project 3 · Illumination-robust HIL-SERL** With a **world model**, decoupled environment appearance from proprioceptive state and re-rendered the data, raising lighting robustness from **65.7%** to over **90%** and recovering to **99%** with light human intervention
- **Project 4 · SuperMimic reproduction with a PPO baseline** Uses a differentiable world model instead of large-scale simulation sampling: BPTT over short imagined rollouts inside the model yields analytic gradients, not PPO-style environment interaction; moved the flagship hand's real2sim2real demo onto it, trained the policy without demonstration data, and benchmarked it against PPO on sample efficiency, convergence speed and final success rate

## RESEARCH EXPERIENCE

**Research on Visuo-Tactile Fusion for In-Hand Dexterous Manipulation** 2025.09 – 2026.01

Master's student · Advisor: Prof. Liu Qian Wireless Multimedia Technology Lab, Dalian University of Technology

- **Paper**: Sole first author in **Frontiers in Robotics and AI** (JCR Q2), 2026.01 — **TouchWGNN: spatio-temporal tactile perception for multimodal dexterous manipulation**
- **Sensor layer**: Designed a piezoresistive tactile sensing array for a five-fingered dexterous hand and its calibration method, turning raw array signals into a tactile representation usable by policy learning
- **Perception layer**: A hierarchical GNN models the finger–contact spatial topology of in-hand touch; a dual-branch high-/low-frequency Transformer captures transient micro-vibrations and steady-state contact forces separately
- **Multimodal fusion**: Cross-attention with frequency-domain gating fuses the asynchronous timing of online visual perception and high-speed tactile servoing — vision supplies the goal, touch supplies the contact state
- **RL and Sim2Real**: Large-scale parallel simulation in Isaac Gym with PPO for in-hand manipulation; grid-searched actuator joint responses for system identification, plus domain randomization to narrow the Sim2Real gap
- **Real-robot deployment**: A real-time tactile acquisition, filtering and force-calibration chain on ROS 2 Humble with a Linux RT kernel for high-frequency control; the policy ran closed-loop on a real five-fingered hand

## SELECTED PROJECTS

**VLA Post-Training with Offline Policy Optimization for Bimanual Garment Folding** 2026.10

- **Task and base model** Folded 4 garment categories (long/short sleeves, trousers/shorts) with tianji bimanual dexterous hands; post-trained the Pi0.5 VLA base with multi-view visual and proprioceptive inputs and action-chunk outputs
- **Offline policy optimization (AWR + RECAP)** The flow-matching head gives only a velocity field with no action density, so log-prob on-policy methods do not apply: AWR reweights data by exp(advantage) under a KL constraint, and RECAP adds dual-mode training — an unconditional pass learns the original distribution, while an advantage-token conditional pass regresses only high-advantage actions and is the sole deployment mode
- **Reward and advantage estimation** Built dense intermediate rewards from keypoint checkpoints, reused the success-probability head as a value function and ran multi-step advantage estimation with GAE, turning sparse success into optimizable per-segment advantages
- **Lightweight world model** Added multi-task auxiliary heads predicting keypoint distances several steps ahead; keypoint-distance sequences replace pixel reconstruction as the dynamics model, giving look-ahead signals at no extra inference cost
- **Distributed asynchronous training pipeline** Decoupled training and rollout nodes via a model/data repository for asynchronous, independently scalable execution; combined domain randomization (texture/lighting/camera/physics), curriculum learning, success replay and hard-example mining
- **Sim2Real and human-in-the-loop** Sim-to-real via behavior-cloning fine-tuning and multi-source data mixing, with camera-pose and speed alignment plus image augmentation; DAgger collected short corrective segments at failures, adding error recovery

## SKILLS & HONORS

**Technical Skills:** `Robot platforms: Newton, ROS 2, MuJoCo, Isaac Gym, LeRobot, MjLab`

`Learning & deployment: PyTorch, Jax, PPO/SAC/RLPD, ACT, Diffusion Policy, VLA, LoRA, Flow Matching, AWR/RECAP, world models, Sim2Real, Sim2Sim`

`Tactile: tactile perception, visuo-tactile fusion, object-centric, GNN`

`Systems: ZeroMQ, WebSocket, Linux RT, shared-memory IPC`

**Languages:** `CET-6`

**Awards:** `Provincial First Prize, Math Competition` `National Third Prize, English Competition`

`Academic Excellence Scholarship` `Merit Student`